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Application of GCN in the JSP problem

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Abstract

One of the biggest problems in computer science is the possibility to make a scheduling for minimize the makespan of the jobs in a system, to optimize it. The problem is well known to be NP-Hard[1], so for finding a good solution we have to follow other possibilities, in this thesis we are going to explore the possibility to do this with a neural network, with the encoding part made as a GNN.

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Chapter 1

Introduction

1.1 Introduction to the JSP (Job Shop Problem)

The Job Shop Problem (JSP) is a variant of the optimal job scheduling[1]. In the optimal job scheduling problem there are n jobs denoted as $J_1, J_2, ..., J_n$ with varying processing time and it's needed to schedule on m machines, while trying to minimize the makespan (it is the total length of the schedule, when all jobs have finished processing)[1].
In this variant each job consists of a set of m operations, which need to be processed in a specific order and in a specific machine. Each operation has a specific machine that needs to be processed and only one job can be processed at a given time.
The difficult part about this problem is that to find the right combination to minimize the makespan you have to search through all the combinations that respects the constraints and this number grows exponentially.
Instead of trying to find the best solution, sometimes a very good solution is enough to work with, this algorithms are called metaheuristic and heuristic methods, for example the dispatching rules (SPT, LPT, FIFO...) or Genetic Algorithms (GA), Local Search, Simulated Annealing...

1.2 Mathematical representation, the MDP

It is possible to represent the JSP problem in a mathematical way, writing it as a Markov Decision Process (MDP)[2]. This approach

let us to apply the Reinforcement Learning to this problem so it is a useful approach.
The MDP is a mathematical model for sequential decision making when the outcome of an action is stochastic, uncertain. All the actions are made by an agent, each action gives a reward by the current state, the action and the state that you are going. The aim is to maximize the expected return of the sum of all rewards.
An MDP is defined by the tuple $M = (S, A, R, T, \gamma, H)$:

- S: State space.
  The state space represents the set of all possible states, can be discrete or continuous.

- A: Action space.
  It represents the set of all possible action that an agent can do in a specific state (different states can have more or less actions). Here too the space can be discrete or continuous.

- R: Reward function.
  The reward function return a number for going from a state to another one by doing an action ($R : S \times A \times S \to \mathbb{R}$)

- T: Transition function.
  It is a function of the probability to go into a certain state by taking an action in a starting state ($T(s'|s, a)$).

- γ: Discount factor.
  It is a number in the range from 0 to 1 (both included) that determines how much importance I should give to the further decisions. $\gamma \in [0, 1]$.

- H: Time horizon.
  It is a natural number that represents the maximum number of actions that can be taken before the process stops.

1.2.1 How to represent a JSP as a MDP

In this section there will be an explanation on how the JSP problem can be written as an MDP

- **State space S:** each state $s \in S$ represents the current partial schedule at a given time step, so:

  - the current availability time of each machine,
  - for each job, the index of the next unscheduled operation (the next operation in its precedence chain)

- **Action space A:** at each state s, the set of available actions $A(s)$ consists of all operations that are eligible to be scheduled next, so operations whose preceding operation in their respective job has been completed, and whose required machine is available.

- **Reward function R:** there isn't a reward given during the episode, the reward is terminal, so after the final step, the sequence is executed and then the reward is given by multiplying with -1 the makespan (a larger makespan is a worse reward).

- **Transition function T:** in this case it is a deterministic function, given a partial list and an action which is chosen an operation, then the next state is going to add the new element to the list.

- **Time horizon H:** it is the number of maximum steps, and in a JSP problem is $n \times m$, the total number of operations

- **Discount factor γ:** the discount factor γ is set to 1 because the reward is only at the end.

Example

Consider a JSP problem 2×2, so for example it is going to have the operations:

$$1^{st} \quad \text{job}: \quad o_{1,1} \rightarrow (M_1, 3) \qquad o_{1,2} \rightarrow (M_2, 4)$$

$$2^{nd} \quad \text{job}: \quad o_{2,1} \rightarrow (M_2, 6) \qquad o_{2,2} \rightarrow (M_1, 1)$$

It is then possible to write this problem as a MDP:

- The initial state s_0 is an empty matrix

- The action space at the beginning, at the state s_0, is $(o_{1,1}, o_{2,1})$ because those are the operations that I can select at the moment

- The transition function is deterministic so the values will be $T([o_{11}]|o_{1,1}, s_0) = 1$, that will represent going to the state where the first operation is the operation $o_{1,1}$ when the precedent state is the initial state s_0 and the action taken is the one of choosing the operation $o_{1,1}$. But it will be set to 0 if the action doesn't correspond to the new state, for example $T([o_{2,1}]|o_{1,1}, s_0) = 0$ or for example $T([o_{1,2}]|o_{1,1}, s_0) = 0$

So if the MDP continues from the precedents points and so it is chosen the operation $o_{1,1}$:

- The new state is now the matrix with one row: $[o_{1,1}]$
- The action space is different from before because in this state the action of choosing the operation $o_{1,1}$ but there is the new possibility of choosing the operation $o_{1,2}$, so it becomes $(o_{1,2}, o_{2,1})$.
- The transition function obviously stay deterministic and changes the value with a 0 or a 1 if the action truly send the previous state in the new state.

1.3 Thesis structure

In a paper of the Department of Computer Science of the University of Turin [3], they approach the creation of a good schedule by using a neural network that will be described in another section. By this idea I decided to try modify the way the problem was seen, making the input a graph to express the problem in a more natural way.
I decided to modify only the encoding part and for now leave the decoder with the same layers as they imagined. The problem is approached with a deep reinforcement technique.
The code is publicly available on GitHub[4].

- **Chapter 2:** a brief recap on how the different arguments explored in this thesis works, starting from the GNN and reaching up to the specific deep reinforcement technique used in the original article and used in my case too.
- **Chapter 3:** an explanation on how I obtained the data, how I modified my data into a graph and how I implemented the models, the one of the paper and the one I modified.

- **Chapter 4:** an explanation and a discussion on the results of the model of the paper and the modified one.

Chapter 2

Background

2.1 GCN

The GCN is a specific type of neural network that takes in input a graph instead of a tensor, how it works in basic neural networks. GCNs were introduced by Kipf and Welling [5].
We will mainly use convolutional layers in order to pass information about the operations. The convolution in this case is a lot of different from the convolution in the CNN, in the CNN the convolution is a weighted sum of the values in a $k \times k$ grid (apply a kernel), instead a GCNN (graph convolutional neural network) does this passages for every node of the graph:

- select all the different nodes connected to the node you are working on
- aggregate all the nodes.
- multiply with the weight matrix, apply a transformation.
- apply an activator function (ex. ReLU).

This operation can be made really fast by doing some matrix multiplications:

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)})$$

The normalization term $\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$ prevents high-degree nodes from dominating the aggregation and ensures numerical stability. With this approach after one layer a node combines the his features and the ones of the nodes directly connected to it, its neighbors. In the second layer a node will have the collection of features that him, its

neighbors and the neighbors of the neighbors and so on.
There is a problem with this approach, it's called oversmoothing, when you apply this convolution too many times the features are too similar and you risk to lose local information.

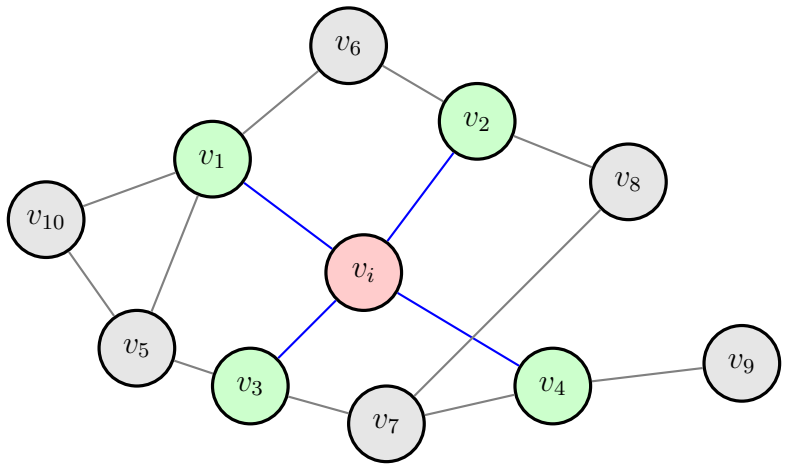

Figure 2.1.1: Message passing in a GCN, the node takes the information from the neighbors.

2.2 LSTM

The LSTM is a specific type of recurrent neural network that help to solve the problem of the vanishing/exploding gradient, a very well known problem in the RNN where the backpropagation of the gradient vanishes/explode because of the recursive layers. LSTMs were introduced by Hochreiter and Schmidhuber [6]. The innovation is that this kind of neurons have a long time memory and a short time memory and their relations.
This kind of layer has 3 different gates: input gate, forget gate and output gate.

- Input gate: it represents the percentage of information that is going to be remembered in the long time memory, so 0 is nothing and 1 is everything.
- Forget gate: it represents how much should be used of the long time memory, how important should be considered.
- Output gate: Decide how to implement the long time memory, the short time memory and the input, at the end is going to

send the final value and that final value is going to be re-used for the short time memory for the next value.

There are 3 values at input that represent: the input, the information in that exact moment, the memory of the previous information (like a short time memory), and a representation of all the previous information (like a long time memory). These values are usually represented as x_t, h_t and c_t.
To decide the values for the different gates we are going to multiply the 3 values x_t, h_{t-1} and c_{t-1} (there is t-1 obviously because we take the memory of previous state) by a value between 0 and 1 we can easily do with a sigmoid function. The input of the sigmoid function is going to be x_t and h_{t-1} and it makes sense, the weight of the long time memory depends on the short time memory and the information I have right now.

- The forget gate is very simple, is just the sigmoid function and we are going to multiply the long time memory c_{t-1} with the sigmoid function, it represents how much information should I use from the old data, how much are important.
- The input gate is a little bit more complicated, there is a sigmoid function but this time does not multiply the long time memory, it is multiplied with a hyperbolic tangent (again the input are x_t and h_{t-1}) and then this multiplication is summed with the long time memory. This operation is like an update of the long time memory with the new information.
- The final gate, the output gate, takes again a sigmoid function of the 2 values but now it multiplies with the hyperbolic tangent of the long time memory c_t, the result of this operation is the new short time memory h_t and the output of the layer.

To summarize, at the beginning I have the forget gate, then after there is the input gate and here I obtain the new updated long time memory c_t, the last one is the output gate and here I use x_t, h_{t-1} and c_t to calculate h_t (that is the output of the node too).
This kind of layer is important to decide which operation we are going to do right now, because it will evaluate the different operation

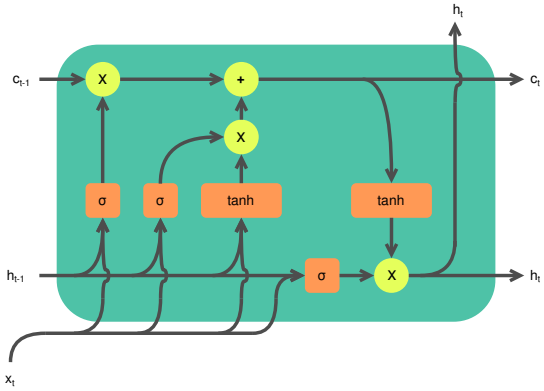

Figure 2.2.2: Image of a LSTM layer

by taking consideration of the previous operations, in the short and in the long time.

2.3 Pointer network

A pointer network is a specific type of neural network where the output is an element of the input, for example if I put a set of elements $\{1, 2, 3, 4, 5\}$ then the Pointer network is going to give an element of the input as output, for example it could be 4. Pointer Networks were introduced in [7].
In this case it is really useful because the neural network can decide which operation to choose in that exact moment and we can do with whatever dimension of the set we want. To do this we are going to repeat the LSTM layer and the pointer mechanism $n \times m$ times.
The main idea of the pointer network is centered in an attention mechanism, in this case also called as a pointer mechanism:

$$\alpha_{t,i} = \text{softmax}(v^T \tanh(W_1 h_i + W_2 s_t))$$

The most critical part to understand is that h_i is a vector that codifies the information about an input, in the paper where the pointer networks are introduced it is done by an LSTM, I decided to do it with a GCN. This means that there will be n vectors h_i where n is the number of inputs.
This mechanism takes a weighted linear sum of the encoding and of the decoding (respectively h_i and s_t) and applies a non-linear transformation, *tanh*, to obtain a value between -1 and 1. To transform this vector into a number we multiply it by the transpose of a vector

v. At the end it is applied a softmax so the final value is between 0 and 1 with a higher value if the value before was high with respect to the other values.
The result of this operation is a vector of values: $\alpha = [\alpha_1, ..., \alpha_n]$ where $\sum_{i=1}^{n} \alpha_i = 1$ (because of the softmax distribution operation).
In the training we are going to select a value randomly according to the distribution but when we are testing or executing the program normally we aren't going to choose randomly but the highest value.

2.4 Deep Reinforcement Learning

The paper decided to apply the reinforcement learning to train the neural network, a policy gradient method.
The policy gradient method is focused to learn the parameters θ for optimizing a stochastic policy π_θ (approximated by the neural network) [8].. The policy takes as argument a state and produce a probability distribution over the possibles actions, and by following the definition of distribution we should have that:

$$\sum_{a \in Act} \pi(a|s) = 1$$

We take those action and by doing this we create an episode τ, more formally: a sequence of states and actions $(s_0, a_0, s_1, a_1, ..., s_H)$ with a return value $R(\tau)$ that is calculated in this way:

$$\sum_{i=0}^{H} R(s_i, a_i, s_{i+1})$$

Also known a finite Horizon Undiscounted Return.
A very important part is the mathematical formula to calculate the probability of going to a certain episode given a fixed policy, given by this formula:

$$P_\theta(\tau) = \rho(s_0) \prod_{t=0}^{H} T(s_{t+1}|s_t, a_t)\pi_\theta(a_t|s_t)$$

Where $\rho(s_0)$ is the probability to have s_0 as first state. Our goal obviously is to have the biggest return as possible, so we are doing a maximization over the expected value of the return:

$$\max_\theta J(\pi_\theta) \quad \text{where} \quad J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$$

We can achieve a good good policy by doing a policy ascent, to do so we need to evaluate $\nabla_\theta J(\pi_\theta)$, the gradient of the expected value of the return of a episode made by following a policy, also known as the policy gradient. It can be shown that the policy gradient is:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau) \right]$$

This is also known as Vanilla policy gradient. The problem is that this method suffers from high variance, given from the fact that $\nabla_\theta J(\pi_\theta)$ is unbiased. To improve we can add into the formula the component of the baseline, a term that depends only from the current state. The formula becomes:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t) \left(\sum_{i=t}^{H} R(s_i, a_i, s_{i+1}) - b(s_t) \right) \right]$$

Chapter 3

Methods applied

3.1 Data

The easiest way to acquire the data is by casually generating them, in this way it's able to generate a lot of different instances with different numbers of jobs and machines.
In my experiments I decided to train only on a 6×6 JSP problem, not too much little but at the same time not too much big. The decision is also based on making easier the training for batches because otherwise there would be the risk to have a batch made of different elements and this would make my GCN raise an exception (there is the possibility to train batches of same dimensions with the dimensions in the batches is different, but in the result section there would be a part where is discussed how my model is able to understand a rule based on any number of n jobs and m machines).

3.1.1 Representation of the data in the article

In the paper, the problem is represented in the easiest possible way: a matrix. The matrix has 4 columns and $n \times m$ rows, each row represents a single operation and it has those feature (in the exact order): the id number of the job i (so from which job comes that operation), the position of the operation in that job j, which machine is required for that operation $M_{i,j}$ and how much time it takes to execute that operation $p_{i,j}$.
So a row is represent like this:

$$o_{i,j} = [i, j, M_{i,j}, p_{i,j}]$$

And the entire JSP problem can be represented as:

$$S^{seq} = \begin{bmatrix} o_{0,0} \\ o_{0,1} \\ \vdots \\ o_{(n-1),(m-1)} \end{bmatrix}$$

In the paper, the goal is to obtain a sequence of instructions where the makespan is as little as possible, so by thinking that at the end the operations will be the same but in a different order the solution can be written as:

$$L^{seq} = PS^{seq}$$

where P is a permutation matrix and L^{seq} is a feasible solution.

3.1.2 Representation of data in my case

I decided to represent the data in a different way, where I can think that it's able to represent my data in a more accurate way: a graph. The graph is structured with 2 category of nodes: operation nodes and machine nodes. The operation node represents a single operation instead a machine node represents a single machine.
The nodes are connected in this way:

- an operation node is connected with another operation node if one is the next one of the other. The connection is unidirectional, from the operation that comes before to the next one.

- an operation node is connected with a machine node if that operation is executed by that machine. This time the connection is bidirectional

Every node has 4 features, but the meaning of the features is different between the 2 nodes:

- in an operation node, the features represent: at which step the operation is in a job (the parameter before called j), the processing time but normalized, the number of operations that are left and the typology of the node (1 if the node is an operation node and 0 if the node is a machine node)

- in a machine node, the features represent: -1 (a value that is useless and impossible for the operation node, so it's a one more check to see if it's a machine node), the total workload in a machine (but it's "normalized" by dividing it with 1000), number of operations that a machine has to execute (it will be n, the number of jobs because it's a JSP problem) and like in the operation node a variable to understand the typology of the node (1 if the node is an operation node and 0 if the node is a machine node).

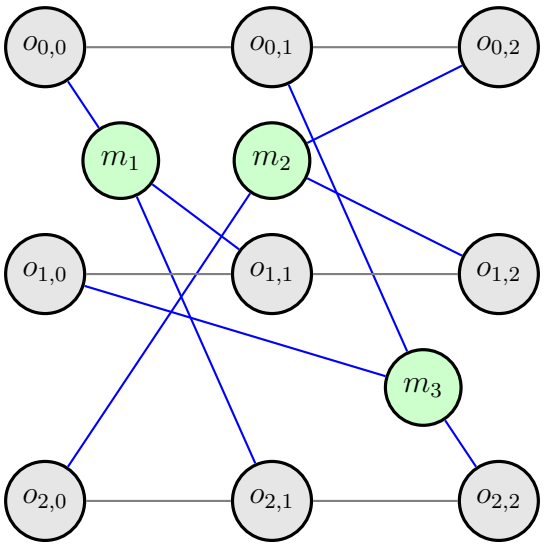

Figure 3.1.1: Example of a 3×3 JSP problem, the gray connections represent the connections between 2 operation nodes (and the are unilateral connection from $o_{i,j}$ to $o_{i,j+1}$ if there exists such operation) and the blue connections represent the connection between a operation node and a machine node.

3.2 Architecture of the model of the paper

The model of the paper is divided in 2 parts, an encoder and a decoder [3]. The encoder is a self-attention-based model where the input is a tridimensional tensor $U \in \mathbb{R}^{N \times (mn) \times 4}$ where m and n are the number of machines and the number of nodes, 4 are the features of an operation and N is the batch size.

3.2.1 Encoder

The first operation of the encoder is to divide the tensor between the first 2 features (the id of the job and the position in that job of the operation) and the other 2 features (which machine should execute that operation and how much time it should take). In each division it's applied a batch normalization and then a linear layer, also known as a dense layer (the most basic type of operation in a neural network). After this then the 2 divisions are summed element by element, it's done another batch normalization and another linear layer (after each linear layer is applied a ReLU as usual).
For the last part it's applied a l multi-head attention layer (they consider $l = 3$), so the final output of the encoder is going to be $H \in \mathbb{R}^{N \times (nm) \times d_h}$ with each row $h_k \in \mathbb{R}^{d_h}$ that are going to be used for the decoder.

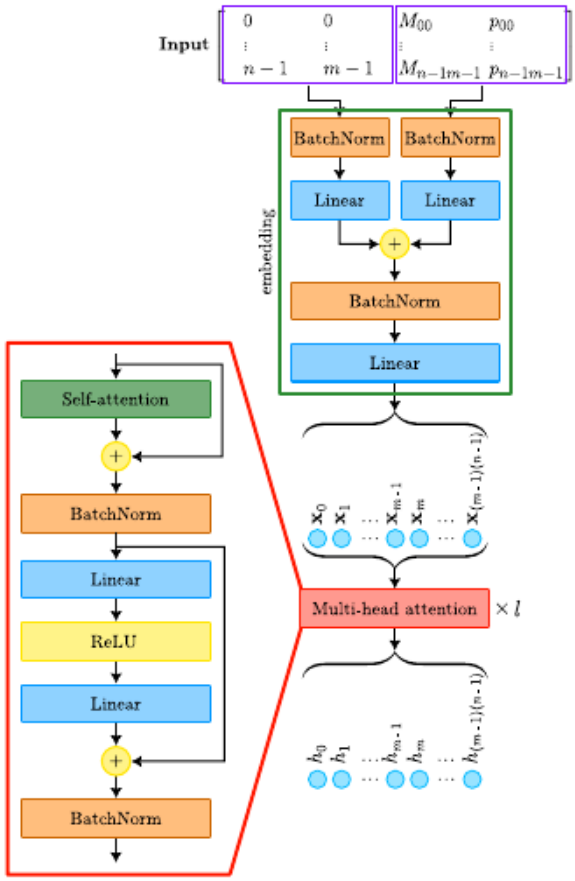

Figure 3.2.2: Image representing the encoder of the paper

3.2.2 Decoder

The decoder has a LSTM layer, very useful in this case because a precedent decision (very far away or not) vastly impact the next choice, after this layer is applied the attention mechanism to make a probability of which element to choose, generating the policy π_θ. The problem is this kind of operation generate a probability over all the values of input, and this isn't a good solution because there is the possibility to choose a operation already been chosen or choose a operation that I should take in the future (rules of the JSP problem). To solve this problem before doing the softmax is going to be applied a mask that would set at $-\infty$ the values so when the softmax distribution is going to be applied the value will be 0. So the final policy distribution will be:

$$\pi_\theta(o'_t|o'_0, o'_1, \ldots, o'_{t-1}, S^{seq}) = \text{softmax}(\text{mask}(u^t|o'_0, o'_1, \ldots, o'_{t-1}))$$

where u^t is the score calculated by the attention mechanism at the t step, mask is the masking procedure applied to obtain possibilities over only operations that are "legals". The policy distribution is obtained $n \times m$ times, so the entire procedure is going to be applied this many times.

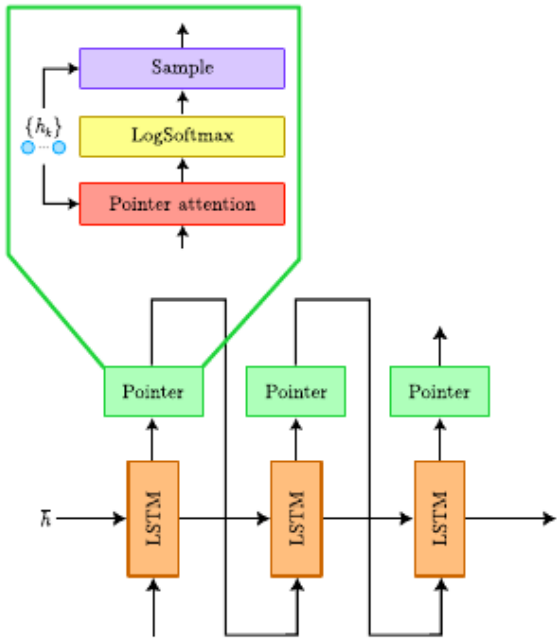

Figure 3.2.3: Image representing the decoder of the paper

Masks

Like it was briefly explained in the previous section, it is necessary to apply a mask. A mask is an array of one and zeros where in this case shows the "legal" possibilities, so if there is False in the position i, then the operation o_i is possible to be selected (because we are going to change values only where the index are True). This is really useful because I have to be able to set some values such that the probability of making a wrong choice is 0, to make so all the impossible decisions are going to be set at $-\infty$, because when it's applied the softmax the value is set to 0, it can be seen in the formula:

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

It is easy to see that the $\lim_{z \to -\infty} e^z = 0$, making the choice impossible to pick.
Are applied 2 kinds of masks to set the values at $-\infty$: one about the sequence of the operations and one about if the operation is already scheduled or not (they are going to be called M^{mask} and M^{sched}).

- M^{mask}: given the instance of a batch k and the index of the operation of the ith job l (also written as $p = i \cdot m + l$), the value M^{mask}_{kp} will be true if $l > j$, where j is the index of the last operation executed by the ith job. This mask is needed to follow the normal flow of the operations, in the right order.

- M^{sched}: given the instance of a batch k and the index of the ith job l (also written as $p = i \cdot m + l$), the value M^{sched}_{kp} will be true if the the jth operation of the ith job has already been scheduled.

The union of this 2 matrices will give in the slots with the value False the operations that it is possible to use (or if it's seen in the other way, the slots where the value are false are the ones that can't be done at least for the moment.

3.3 Architecture of my model

In my model I'm going to change only the encoding part and leave the decoder the same as the one in the paper.
My encoder is based on a graph representation of the data, so it will be a GNN, more specifically a GCN. The model start with a GCN layer and and then it has 2 blocks of layers made as: Layer Normalization, ReLU activators and Graph's Convolution. It is utilized only a total of 3 graph convolutional layers to prevent the risk of oversmoothing.
To apply the decoder of the paper the output should be a tensor $H \in \mathbb{R}^{N \times (nm) \times d_h}$, so the operation that is done to obtain this is a filter of the nodes to obtain only the operation nodes and then translate the graph into the tensor.

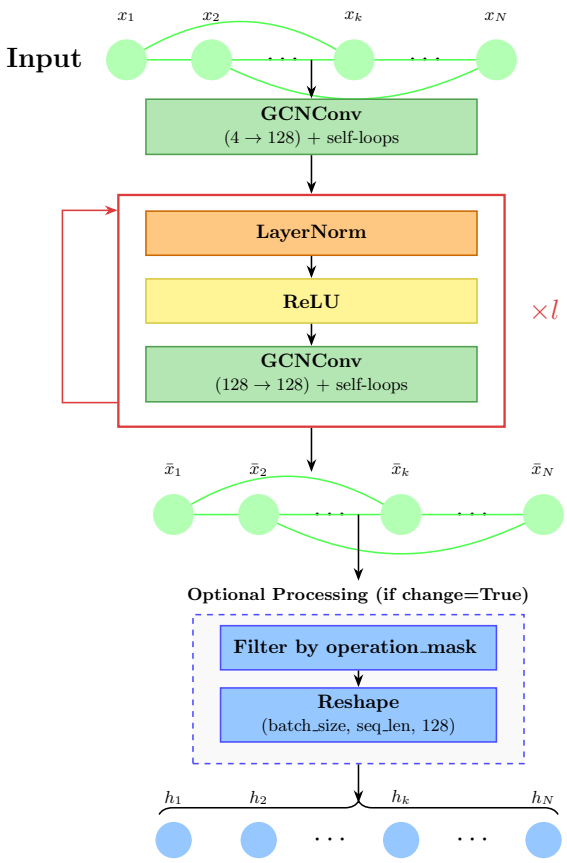

Figure 3.3.4: Image representing the encoder that I wrote where there is at first a GCN Layer and then an arbitrary number of LayerNorm, Relu and GCN Layers that are repeated in this order, in this case $l = 2$.

3.4 Training of the neural networks

The models were trained using a policy gradient method (with an ADAM optimizer):

$$\nabla_\theta L(\pi_\theta) = \mathbb{E}\left[(C_{max}(L^{seq}) - b(S^{seq}))\nabla_\theta \log P_\theta(L^{seq}|S^{seq})\right]$$

where $P_\theta(L^{seq}|S^{seq}) = \Pi_{t=0}^{n \cdot m-1} \pi_\theta(o'_t|o'_0, o'_1, ..., o'_{t-1}, S^{seq})$ so it's the probability of getting the solution L^{seq} following the policy π_θ and given a problem S^{seq}. After each epoch, the baseline is updated with the optimized policy's weights only if the policy is statistically better. This update ensures that we are training our policy with the the best policy found so far.
The models were trained 150 epochs, with 10 thousand elements as a training set and a thousand elements for validation and test set. They were trained in CPU and not GPU because the model hasn't too many variables to calculate, it takes more time to send the data into the GPU than calculating everything in the CPU.

Chapter 4

Results and comparison

These models where trained only using a 6×6 JSP problem, in the paper they trained their model with different settings to make the neural network understand how to solve the problem using different numbers of n jobs and m machines.
In this work the models were trained with only the 6×6 JSP problem to test an hypothesis: if a graph representation and so a GNN as an encoder is able to make the neural network understand the problem with a different number of n jobs and m machines.

4.1 6×6 JSP problem

The model of the paper is making on average a policy where the result is just 23% more than the actual best solution. The model with the GCN encoder instead gives a solution that is only 19% more than the actual best solution.
The models reached an equilibrium at around 25-30 epochs (both of them), we can notice that the graph representation reaches a better result from the beginning, so it's without any doubt that the GCN encoder is able to understand the dynamic of the problem in a more accurate way than the matrix representation.

This implies that the graph representation is able to express the problem and its characteristics in a more accurate way. The results in the image are obtained through a test set of 1000 elements.

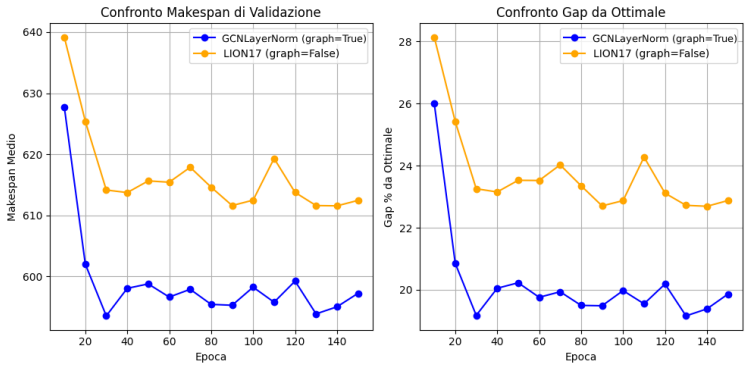

Figure 4.1.1: Image representing the results of the validation test on the models on the 6×6 JSP problem.

4.2 n × m JSP problem

After the results obtained through the training of the model with only the 6×6 JSP problem both the models were tested with different sizes of the JSP problem, to see if at least one of them is able to understand the logic with different number of jobs and machines.
In the paper, the normal model was trained with different sizes problems, so it was reasonable to expect that the model wouldn't be able to understand this complicate feature on its own with only one kind of problem.
The only hope was the modified version with the graph, hoping that it was able to understand how it should work with, hoping that the graph is able to express some constraints in a different way such that it is easier to find a solution even if the number of connections and nodes vary. It was reasonable to think it because the model is just a message passing model, so it just make a weighted sum of the values in different nodes.

Mostly of the cases it outperformed the model of the paper, but at the beginning of the test it seemed as the model of the paper was able to make a better schedule when the number of the number of the machines was higher than the number of the jobs, for example in the 5×10 JSP:
This hypothesis is contradicted by the results on the 6×15 JSP problem where the model with the GCN encoder was able to make a better schedule than the other one.

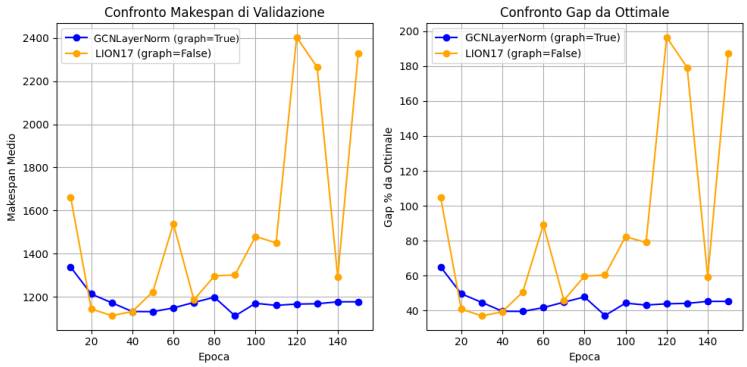

Figure 4.2.2: Image representing the results of the validation test on the models on the 10×10 JSP problem.

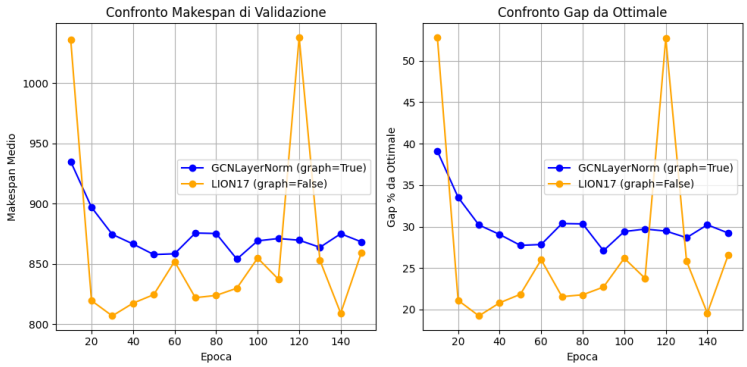

Figure 4.2.3: Image representing the results of the validation test on the models on the 5×10 JSP problem.

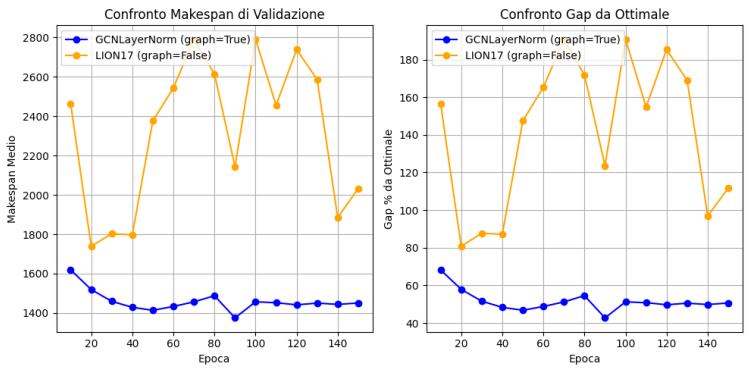

Figure 4.2.4: Image representing the results of the validation test on the models on the 6×15 JSP problem.

There is a strong suggestion that the GCN encoder is able to understand the properties of the JSP problem through the graph representation for 3 reasons: to the fact that the model outperform most of the time the model of the paper, the model doesn't spike randomly like it does the model of the paper for all the tests different from the 6×6 problem and on all different sizes the GCN encoder model follows more or less the same path instead the model of the paper follows more random paths, it doesn't seem to follow a pattern. This fact is truly useful because it means that it is possible to train the model on samples of little dimensions and obtain a very good result that works very well even for very large size even if it hasn't ever seen one.
The fact that both models were trained only on 6×6 JSP problems is another element in favor of considering the baseline's superiority on the 5×10 problem as just an exception, because it doesn't make sense that it would generalize better to this specific configuration when it has only ever seen 6×6 JSP problems.
The models were tested with different sizes too, everything represented on a table, for simplicity

JSP problem size	Perfect schedule	GCN encoder model	Paper model
6×6	55	65	68
10×10	930	1218	1320
5×10	545	817	712
6×15	545	712	850
10×5	622	765	778
30×20	4545	5000	22500

4.3 Possible explanation of the performance of the model

The fact that the model with the GCN encoder is able to make a good schedule for every kind of $n \times m$ JSP problem is probably due to the fact that the topology of the graph gives a good path to the neural network to follow, it is able to passage the right information and it probably helps the fact to have added the machine nodes, they probably works as a collector of information through the different

nodes and so it isn't needed a very deep network.

4.4 Future works

There could be a lot of new possibilities to improve this kind of architecture, from modifying the encoder into a GAT, GraphSAGE and many other kinds of graph neural network, making more attention to the machine nodes. Another possibility is to work on the decoder, for example the choice of the operations are made with a very little attention mechanism after a LSTM layer, instead could be applied a GRU layer or a Transformer. Another possibility to the decoder is to implement some kind of layers that let the decoder works with a graph, another graph neural network to preserve the information and structure of a graph, a Graph Pointer Network. In this thesis the network isn't so much deep, probably with a lot of more computational power there is the possibility to make it much deeper, or it can be trained the neural network with different kind of problems, much bigger with the one I considered, to see if the model of the paper is also able to work very well with different size problems after the training with some diversity of problems. Another interesting possibility is to implement this work so it is dynamic, it changes the list of operations based to the fact if someone add a new job in the middle of the execution of the operations.

Glossary

JSP section

$\mathbf{S_{seq}}$: Sequential representation of the JSP problem in a matrix form.

$\mathbf{S_{seq}}$: Sequential representation of the solution of a JSP problem in a matrix form.

$\mathbf{o_{i,j}}$: Vector representing the jth operation of the ith job. It is made by 4 elements: $[i \quad j \quad p_{i,j} \quad M_{i,j}]$

$\mathbf{p_{i,j}}$: Natural number representing the time that takes the machine to execute the jth operation of the ith job.

$\mathbf{M_{i,j}}$: Natural number representing the machine that should execute the jth operation of the ith job.

GCN section

$\mathbf{G = (V,E)}$: Graph, represented by a set of vertices $\mathbf{V}$ and a set of edges that connect 2 different vertices $\mathbf{E} = \mathbf{V} \times \mathbf{V}$.

$\mathbf{A}$: Adjacency matrix of a graph, it represents the various connections (edges) of a graph.

$\tilde{\mathbf{A}}$: Adjacency matrix of the graph plus all the self-loops, so each node is connected to himself.

$\mathbf{H^{(l)}}$: Feature matrix $\in \mathbb{R}^{N \times N_f}$ where N is the number of nodes and N_f is the number of features in a node (a row is a node and a column is a feature) at the l-th layer.

$\mathbf{W^{(l)}}$: Weight Matrix at the l-th layer.

$\tilde{\mathbf{D}}$: Diagonal Matrix of the degrees of a node, it represents the number of connection a node has so: $D_{ii} = \sum_{j=1}^{n} \tilde{A}_{ij}$ and $D_{ij} = 0 \ \forall i \neq j$

$\boldsymbol{\sigma}$: Activation function, usually the ReLU

LSTM section

$\mathbf{x_t}$: Input at time t.

$\mathbf{h_t}$: Short time memory of the cell LSTM.

$\mathbf{c_t}$: Long time memory of the cell LSTM.

RL section

$\mathbb{E}$: Expected value of a distribution function.

$\mathbf{T}(\mathbf{s_{t+1}}|\mathbf{s_t}, \mathbf{a_t})$: Function representing the transition from the state $\mathbf{s_t}$ to the state $\mathbf{s_{t+1}}$ by making the action $\mathbf{a_t}$.

$\boldsymbol{\nabla}$: Represent the gradient of a multivariable function.

$\boldsymbol{\pi_\theta}$: Policy based on the parameter θ

Acronym

JSP : Job Shop Problem

GNN : Graph Neural Network

GCN : Graph Convolutional Network

LSTM : Long Short Time Memory

MDP : Markov Decision Process

NP-Hard : Non-deterministic Polynomial-time Hard

RL : Reinforcement Learning

DRL : Deep Reinforcement Learning

GA : Genetic Algorithm

SPT : Shortest Processing Time

FIFO : First In First Out

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Appendix A

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